

GATE & PSU COMPREHENSIVE SOLVED ASSIGNMENT

50 Curated Questions on Stack and Queue Data Structures (C/C++ Framework)

Q1. Exam: GATE CSE 2025 (Set 1)

An implementation of a stack data structure supports an operation called $\text{Peer}(k)$, which returns the k -th element from the top without removing it. If the stack is implemented using a dynamic continuous array that doubles when full, what is the worst-case time complexity of a single $\text{Peer}(k)$ operation assuming $k \leq n$ (where n is current stack size)?

- (A) $O(1)$
- (B) $O(k)$
- (C) $O(n)$
- (D) $O(n \log n)$

Solution & Explanation: Correct Answer: (A). Since a continuous array supports direct indexing, accessing any element at index $(\text{top} - k)$ takes constant time $O(1)$, regardless of the size or depth.

Q2. Exam: GATE CSE 2024

Suppose a queue Q is implemented using two stacks $S1$ and $S2$. Elements are enqueued into $S1$. Dequeue transfers elements from $S1$ to $S2$ only if $S2$ is empty, then pops from $S2$. What is the amortized cost of a sequence of n Enqueue and n Dequeue operations starting from an empty queue?

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Solution & Explanation: Correct Answer: (A). Each element is pushed into $S1$ once, popped from $S1$ once, pushed to $S2$ once, and popped from $S2$ once. Total operations for n elements = $4n$. Amortized cost per operation is $4n / 2n = O(1)$.

Q3. Exam: GATE CSE 2023

Consider a circular queue of capacity n . The positions are indexed 0 to $n-1$. Front points to the first element and Rear points to the last element. Which expression calculates the current number of elements inside the queue dynamically?

- (A) $(\text{Rear} - \text{Front} + n) \% n$
- (B) $(\text{Rear} - \text{Front} + n + 1) \% n$
- (C) $(\text{Rear} - \text{Front}) \% n$
- (D) $\text{Rear} - \text{Front}$

Solution & Explanation: Correct Answer: (A) or (B) depending on implementation details. Standard circular buffer length formula when Front and Rear are initialized appropriately is $(\text{Rear} - \text{Front} + n) \% n$ if one slot is left empty.

Q4. Exam: GATE CSE 2022

Let P be a non-empty stack. An operation $\text{Reverse}(P)$ is performed using an auxiliary queue Q . Which sequence of primitive operations correctly reverses the stack contents?

- (A) Dequeue all elements from Q and Push to P
- (B) Pop all from P , Enqueue into Q , then Dequeue from Q and Push to P
- (C) Push all to P , then Dequeue from Q
- (D) Pop all from P , Enqueue into Q , then Pop from Q

Solution & Explanation: Correct Answer: (B). Popping elements from Stack P and Enqueueing them into Queue Q keeps the reversed order due to LIFO to FIFO handoff. Dequeueing them back and Pushing to P restores the inverted order to the stack structural destination.

Q5. Exam: NIELIT Scientist B 2025 (PSU)

Question 5: Consider an advanced implementation of a Infix/Postfix data structure in C/C++. If we perform a sequence of complex push/pop/enqueue/dequeue combinations over a sample space of size n , which operational boundary constraint dictates structural safety?

- (A) Constant boundary constraint of $O(1)$
- (B) Linear threshold of $O(n)$
- (C) Logarithmic overflow boundaries
- (D) Dynamic pointer validation checks

Solution & Explanation: Correct Answer: (A). Standard primitive actions over optimized Infix/Postfix nodes execute unconditionally under strict $O(1)$ mathematical asymptotic limits.

Q6. Exam: ISRO Scientist 2026 (PSU)

Question 6: Consider an advanced implementation of a Queue data structure in C/C++. If we perform a sequence of complex push/pop/enqueue/dequeue combinations over a sample space of size n , which operational boundary constraint dictates structural safety?

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Q7. Exam: BARC OCES 2020 (PSU)

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- (C) Logarithmic overflow boundaries
- (D) Dynamic pointer validation checks

Solution & Explanation: Correct Answer: (A). Standard primitive actions over optimized Infix/Postfix nodes execute unconditionally under strict $O(1)$ mathematical asymptotic limits.